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Methods

DNAreader: accurate prediction of DNA-binding residues in structured and disordered proteins using transformers and contrastive learning

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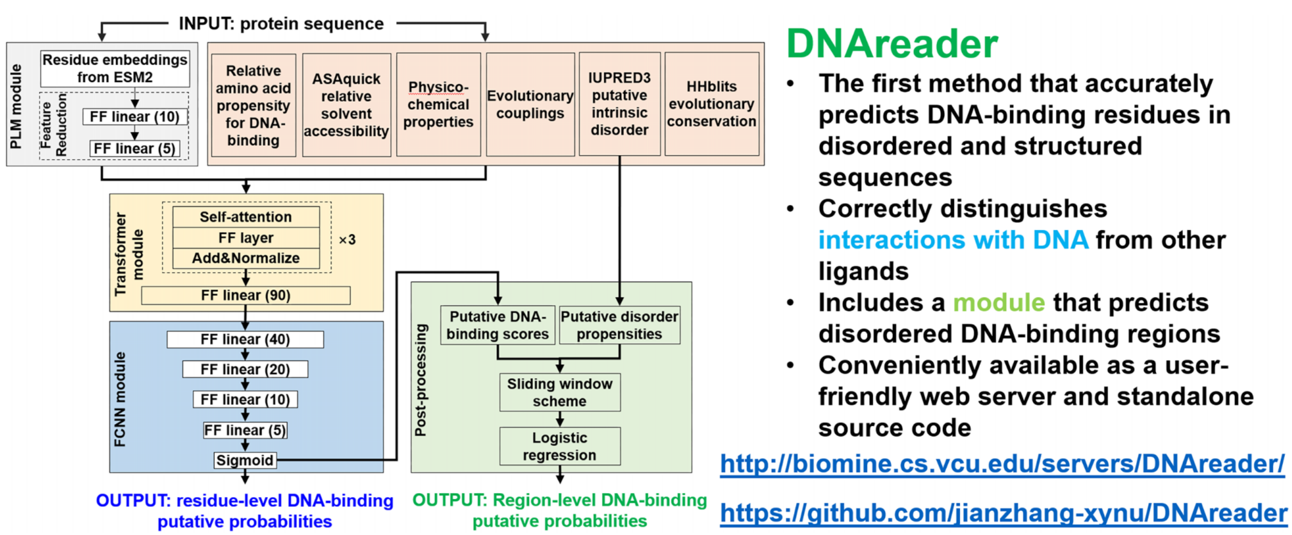
Abstract

Accurate predictions of DNA-binding residues (DBRs) in protein sequences facilitate decoding molecular-level mechanisms underlying cellular functions that involve protein–DNA interactions. While dozens of these predictors have been released, they target either structured or intrinsically disordered regions (IDRs), and the latter were trained to predict less detailed DNA-binding IDRs rather than DBRs. Given this dichotomy, the structure-trained methods underperform on disordered proteins, and vice versa. Moreover, they suffer from high cross-prediction rates, incorrectly labeling many residues that interact with non-DNA ligands as DBRs. We address these issues by introducing DNAreader, the first predictor specifically designed to predict DBRs in the structured and disordered sequence regions. DNAreader relies on an innovative stacked transformer encoder network that combines batch training and contrastive learning, which substantially boosts predictive performance. Using two low-similarity test datasets, we demonstrate that DNAreader statistically outperforms existing tools, performs well for structured and disordered regions, and produces very few cross-predictions. We also developed the DNAreader$_{DBIDR}$ module, which accurately predicts DNA-binding IDRs, providing flexibility to identify DBRs within IDRs or to predict entire disordered DNA-binding regions. We release DNAreader as a user-friendly web server at http://biomine.cs.vcu.edu/servers/DNAreader/, with the corresponding source code at https://github.com/jianzhang-xynu/DNAreader.

Graphical abstract



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Introduction

Protein–DNA interactions drive numerous cellular functions, such as regulation of gene expression, chromatin remodeling, and DNA replication, repair, and recombination [1, 2]. Knowledge of the DNA-binding residues (DBRs) in protein sequences facilitates the identification of DNA-binding proteins and the uncovering of molecular mechanisms that underpin their cellular functions. Several experimental techniques, such as X-ray crystallography, NMR spectroscopy, and cryo-electron microscopy, are employed to structurally characterize these interactions [3], thereby indirectly providing annotations of DBRs [4, 5]. However, these molecular-level details are known for only a small fraction of the DNA-binding proteins, with ∼9200 protein–DNA complex structures in the worldwide protein structure repository, the Protein Data Bank (PDB) [6, 7]. Computational methods provide a cost-efficient, high-throughput approach to generate putative annotations of DBRs in protein sequences [8–13]. They are trained/designed using limited experimental data, and if tested to be accurate on a set-aside test data, they can be used to predict DBRs for the vast majority of protein sequences that remain uncharacterized. Based on a manual analysis of relevant PubMed queries and the content of recent surveys [8–11], we identified ∼50 studies that reported sequence-based predictors of DBRs over the last two decades [13–61]. We summarize them in Supplementary Table S1 and observe considerable recent growth in this area, with seven new predictors published since 2024 [51–55, 60, 61]. Each method belongs to one of the two distinct categories: structure-trained versus disorder-trained. The structure-trained methods were trained on DBR annotations from structured protein–DNA complexes, typically collected from the PDB [4, 5, 7]. The disorder-trained techniques rely on DNA-binding annotations for intrinsically disordered regions (IDRs) [62, 63], obtained from the DisProt database [62]. We recently focused on the dichotomy of these annotations and the fact that the corresponding predictors perform relatively poorly for the opposite annotation type [9, 61–64]. Structure-based annotations typically reflect binding within structured pockets on the protein surface. In contrast, the disorder-based annotations are characterized by the absence of a stable tertiary structure for the binding pocket in the apo state, which undergoes binding-induced folding and which may assume different conformations depending on the specific ligand that interacts with a given disordered sequence region [65–67]. Significantly, binding annotations differ between the two data sources. Structure-based sources annotate DBRs, i.e., amino acids in proximity to DNA that are assumed to interact directly with it. In contrast, disorder-based sources annotate DNA-binding IDRs, i.e. disordered sequence regions that interact with DNA and which consist of at least 10 consecutive amino acids. Correspondingly, structure-trained methods predict DBRs, whereas disorder-trained tools predict DNA-binding IDRs. The poor predictive performance across these annotations can be attributed to the fact that these predictors were trained on one annotation type and not evaluated on the other during their design. Supplementary Table S1 reveals that most of these methods are structure-trained (42 out of 47), with only 4 disorder-trained tools.

Although disorder-trained methods were designed to predict DNA-binding IDRs, only some residues in these regions participate in DNA interactions, while the remaining amino acids merely reside in the same IDR. In other words, the two predictor types were trained to yield inconsistent predictions of binding residues in structure-annotated interactions versus binding regions in disorder-annotated interactions (Supplementary Table S1). In 2024, we published the HybridDBRpred meta-predictor that generates relatively accurate results across the two annotation types [61]. However, this solution merely combines results from selected structure- and disorder-trained methods, falls short of improving predictive performance, and was designed to produce inconsistent predictions of binding residues versus regions. Moreover, many of the existing tools suffer high levels of cross-predictions [47, 52, 61, 64], defined as predictions of interactions with an incorrect partner molecule type [68], i.e., predictors of DBRs also predict residues that interact with proteins and RNA as DBRs. This is a significant issue, essentially rendering their predictions ligand-agnostic. This stems from using training data that did not cover proteins that interact with non-DNA partners, and from predictive models that have not been optimized to minimize predictions of interactions with non-DNA partners. Supplementary Table S1 shows that the development process for only 8 of the 47 methods involved quantifying cross-predictions. Furthermore, Supplementary Table S1 indicates that 10 methods exclusively produce binary predictions (residues are predicted as either DBRs or non-DBRs). At the same time, it is preferable to predict real-valued scores that quantify the propensity for DNA binding, yielding more granular and informative predictions.

Motivated by the above discussion, we conceptualized, designed, comparatively evaluated, and deployed a universal and accurate predictor of DBRs in protein sequences, DNAreader (predictor of **DNA** binding disor**d**ered **a**nd structu**re**d residues). This is the first tool designed to predict DBRs for both structured and disordered protein–DNA interactions. Moreover, it generates real-valued propensities and achieves low cross-prediction rates. DNAreader includes an additional module that we designed to predict DNA-binding IDRs, aligning with the scope of current disorder-trained tools and the underlying annotations in DisProt. We constructed a new dataset incorporating structure- and disorder-annotated proteins from the PDB and DisProt, covering annotations of DBRs and residues that interact with other ligand types. The latter allowed us to quantify and minimize the cross-predictions. DNAreader relies on a modern transformer encoder network and protein language model (PLM), which we combined with two innovations that involve designing batch training [69] and contrastive learning [70] approaches to maximize predictive performance and minimize cross-predictions. We released DNAreader as a web server at http://biomine.cs.vcu.edu/servers/DNAreader/, where we also provided the underlying training, validation, and test datasets. We also provide the source code at https://github.com/jianzhang-xynu/DNAreader. In a nutshell, DNAreader solves issues of inconsistent annotations and cross-predictions with a cutting-edge deep learning model that accurately predicts DBRs, with the added flexibility to predict DNA-binding IDRs.

Materials and methods

Selection of methods for comparative evaluation

We comparatively evaluated DNAreader against a representative collection of existing predictors of DBRs, which we selected based on the following three criteria: (i) must be available to make predictions (i.e. has working web servers or code that can be successfully installed and run locally); (ii) must be computationally efficient to make predictions on our large test dataset that covers 437 proteins (i.e. predicts an average size protein with 300 residues in under 10 min); and (iii) must generate real-valued propensity scores, which are required to compute various metrics of predictive performance. Correspondingly, we included eight structure-trained methods (chronologically): BindN+ [71], TargetS [29], TargetDNA [33], DNAPred [40], NCBRPred [43], DNAgenie [45], iDRNA-ITF [72], and CLAPE [55]; two disorder-trained methods: DisoRDPbind [56] and DeepDISObind [59]; and the sole structure and disorder-trained meta-predictor, HybridDBRpred [61], which combines results produced by the structure-trained DNAgenie and DNAPred, and the disorder-trained DisoRDPbind. We also include a popular, recently released predictor of protein and peptide binding IDRs, ANCHOR2 [73] (i.e., a predictor of binding residues interacting with a different ligand type), to provide a baseline result. This tool should produce low accuracy (i.e., it should not be able to predict DBRs accurately) and high cross-predictions, since some of these correspond to correct predictions of protein/peptide binding residues. Notably, 6 of the 12 selected methods, highlighted in bold in Supplementary Table S1, were published within the last 5 years (NCBRPred, DNAgenie, iDRNA-ITF, DeepDISObind, CLAPE, and HybridDBRpred), ensuring that modern predictors are well represented.

The method with the most similar scope to DNAreader is HybridDBRpred, which is a simple meta-predictor that combines the outputs of the structure-trained and disorder-trained DBR predictors released prior to 2023. We implemented an updated variant of HybridDBRpred, using the same approach and incorporating the best structure-trained and disorder-trained predictors, including methods published after 2023. Correspondingly, the resulting HybridDBRpred2 uses the same predictive architecture as HybridDBRpred and combines predictions from the disorder-trained DisoRDPbind [56, 57] and the structure-trained CLAPE [55] and iDRNA-ITF [47]. We also implemented a simple baseline meta-predictor that averages the normalized predictions of all 10 structure-trained and disorder-trained predictors, which we refer to as Meta-average. Altogether, we compared DNAreader against 14 methods, including 8 structure-trained predictors, 2 disorder-trained predictors, 3 meta-predictors (HybridDBRpred, HybridDBRpred2, and Meta-average), and the baseline predictor of protein/peptide-binding IDRs, ANCHOR2.

Datasets

We aimed to obtain a large collection of structure- and disorder-annotated DNA-binding proteins, including annotations of DBRs for both protein types and DNA-binding IDRs for the disorder-annotated proteins. These proteins should also interact with other ligand types, providing sufficient annotations of the corresponding binding residues, which we used to quantify and minimize cross-predictions. We sourced these data from our recently published study that developed the HybridDBRpred meta-predictor [61]. This study included training, validation, and test datasets comprised of structure-annotated and disorder-annotated proteins. However, the disorder-annotated proteins in these datasets lack annotations for DBRs and cover only DNA-binding IDRs. Correspondingly, we expanded these datasets with disordered proteins that have annotations of DBRs, which we obtained through a three-step process. First, we collected 692 proteins with experimentally annotated linear interacting peptide (LIP) regions and IDRs from MobiDB [74] and DisProt [75]; 503 of these proteins contained information on interacting residues in MobiDB. Second, we identified 62 proteins with at least one DBR within either IDR or LIP region, corresponding to DNA-binding IDRs with annotated DBRs. Third, we annotated residues that interact with other, non-DNA ligands for these 62 proteins using experimental annotations from MobiDB. Since the study from Zhang *et al.* [61] ensured that the test proteins share <25% sequence similarity with the training and validation proteins from that study, we divided the 62 proteins into two subsets: one that we merged with the test proteins from Zhang *et al.* [61] and the other, which we merged with the training and validation proteins from Zhang *et al.* [61] and which shares <25% similarity with the merged set of the test proteins. To properly account for the constraints on the similarity between training and test data, we clustered sequences of all 62 proteins and DNA-binding IDRs with the test proteins from Zhang *et al.* [61] at 25% similarity using BLASTclust, and we merged proteins in clusters that exclude the test proteins and their DNA-binding IDRs with the training and validation proteins from Zhang *et al.* [61]. We combined the remaining proteins with the test proteins from Zhang *et al.* [61]. This way, the combined collection of test proteins and their DNA-binding IDRs shares <25% sequence similarity with the combined collections of the training and validation proteins.

DNAreader includes three predictive models: (i) a feed-forward neural network that converts outputs of the ESM-2 PLM to inputs for the DBR predictor; (ii) a transformer encoder-based model that produces predictions of DBRs; and (iii) a regression module that uses these transformer-generated outputs to predict DNA-binding IDRs. The first two models aim to predict DBRs; thus, they should be designed using proteins with these annotations. Correspondingly, we clustered the training dataset from Zhang *et al.* [61] together with the new disordered proteins with DBR annotations that are dissimilar to the test proteins using BLASTclust at 25% similarity to develop training and validation datasets for the development of these two models. We divided the resulting clusters into training dataset 1A, training dataset 1B, and validation dataset 1, comprising ∼25%, 50%, and 25% of the clustered proteins, respectively. We utilized training dataset 1A and validation dataset 1 to train and optimize model 1 (the feed-forward neural network). We employed the training dataset 1B and validation dataset 1 to design and optimize the transformer encoder-based network. The third regression-based model aims to predict DNA-binding IDRs; therefore, it does not require disordered proteins with DBR annotations for training. Correspondingly, we clustered the validation dataset from Zhang *et al.* [61] at a 25% similarity threshold, assigning half of the clusters to create training dataset 2 and the other half to create validation dataset 2, which we used to develop the regression model. Clustering ensures that proteins in the validation datasets share <25% similarity with their corresponding training proteins. This allows us to minimize potential overfitting to the training data, as we assess the training progress using results on the corresponding (and dissimilar) validation dataset.

We comparatively evaluated DNAreader against a representative set of 14 methods using a collection of blind (low-similarity) test proteins. We used two test datasets: test set 1, which is compatible with past test datasets that include DBRs for the structure-annotated test proteins and DNA-binding IDRs for the disorder-annotated test proteins, and test set 2, which contains DBRs for both structure- and disorder-annotated test proteins. We sourced the test set 1 from the test dataset in Zhang *et al.* [61]. We processed this test set to exclude proteins similar to the training datasets used to train the 14 methods; these test proteins already share <25% similarity with the training and validation proteins used to train DNAreader. To this end, we clustered the combined collection of the test proteins from Zhang *et al.* [61] and training proteins for these methods using BLASTclust at 25% similarity, and we selected proteins from the clusters that excluded the training proteins. The resulting test set 1 includes 430 proteins. Test set 2 contains the new disordered proteins with DBR annotations that we collected, which share <25% similarity with the training and validation proteins. We combined them with the disordered non-DNA-binding proteins and structured proteins from test dataset 1, in proportions mirroring those of structured versus disordered proteins in test dataset 1, for a total of 231 proteins. In a nutshell, test set 1 (DBRs for the structure-annotated proteins and DNA binding IDRs for the disorder-annotated proteins) and test set 2 (which covers DBRs for both structure-and disorder-annotated proteins) share low similarity with training and validation proteins, and they have similar proportions of disordered, structured, and binding residues, allowing us to compare results across them directly. We detail the training, validation, and test datasets in Supplementary Table S2. These datasets, including annotations of DBRs and DNA-binding IDRs, are provided at http://biomine.cs.vcu.edu/servers/DNAreader.

Assessment metrics and statistical analysis

DNAreader and the selected 14 predictors and meta-predictors output real-valued propensities for DNA binding. We evaluated these scores with the popular area under the ROC curve (AUROC) metric. Since DBRs account for a small fraction of residues, we also computed the AULCratio, which was applied in several related studies [13, 35, 45, 61]. AULCratio quantifies AUC for part of the ROC curve where the number of predicted DBRs does not exceed the actual number of DBRs, which is divided by the corresponding AUC of a random predictor. Thus, an AULCratio of 1 corresponds to the random level prediction, while higher AULCratio values express the rate of improvement over the random result.

The binary predictions (DBR versus non-DBR) are generated from the propensities by applying a threshold: residues with propensities exceeding the threshold are predicted as DBRs, while those with propensities below it are classified as non-DBRs. We calibrate thresholds across predictors to facilitate side-by-side comparisons. We utilize multiple threshold calibrations to evaluate predictors under different scenarios, including thresholds that generate low false positive rates of 0.1 and 0.2, which are motivated by the comparably low fraction of DBRs, as well as thresholds that produce sensitivity rates of 0.5 and 0.7, which quantify scenarios where large fractions of DBRs are correctly predicted. Under these four calibrations, we assess the binary predictions utilizing several popular metrics that include sensitivity (fraction of correctly predicted native DBRs), specificity (fraction of correctly predicted native non-DBRs), and the F1-score (harmonic mean of precision and sensitivity, which is particularly suitable for the class-imbalanced scenarios, like this prediction).

Motivated by several recent studies [45, 47, 52, 61, 64, 76], we also analyzed false positives (non-DBRs incorrectly predicted as DBRs), which are categorized into two subtypes: cross-predictions (residues that bind other ligand types incorrectly predicted as DBRs) and over-predictions (residues that do not bind ligands incorrectly predicted as DBRs). Correspondingly, we computed the cross-prediction ratio (CPRratio) and over-prediction ratio (OPRratio), defined as the cross-prediction rate (fraction of residues that interact with non-DNA ligands predicted as DBRs) and the over-prediction rate (fraction of non-binding residues incorrectly identified as DBRs) of a random predictor divided by the rate of a given method, respectively, to evaluate the binary predictions. CPRratio or OPRratio values quantify the degree of improvement over a random predictor, where a value ≤ 1 suggests that a given predictor is equivalent to or worse than a random result. Analogous to the ROC curve, we assessed the real-valued propensities using the cross-prediction and over-prediction curves, which quantify the cross-prediction and over-prediction rates relative to the true positive rates. We computed the area under the cross-prediction curve (AUCPC) and the area under the over-prediction curve (AUOPC). Lower AUOPC and AUCPC values indicate that the corresponding method yields smaller over-predictions and cross-prediction rates, respectively.

We assessed the statistical significance of differences in predictive performance between the DNAreader and other predictors. These tests investigate whether the differences are consistent across different datasets. To this end, we performed 100 evaluations on randomly selected subsets of 50% of the test proteins. We compared the corresponding 100 results using the Student's *t*-test, assuming these data were normally distributed (as evaluated with the Anderson–Darling test at a 0.01 significance level). Otherwise, we utilized the Wilcoxon rank-sum test. We considered differences statistically significant if the resulting P-value <.01.

DNAreader

We developed DNAreader with two primary objectives: to generate accurate predictions of DBRs across structure- and disorder-annotated proteins and to minimize cross-predictions for residues that bind other types of ligands. Correspondingly, DNAreader relies on modern transformer encoder networks and transfer learning to boost predictive performance. We also creatively fused contrastive learning and batch training to maximize predictive quality and address cross-predictions. The DNAreader's workflow is shown in Fig. 1. The design involves three steps: (i) encoding inputs for the transformer network; (ii) batch training the network; and (iii) applying outputs of this network to predict DNA-binding IDRs (DBIDRs). In the first step (Fig. 1A), we encode the input protein sequence into a sequence profile by computing residue embeddings and sequence-derived features. We obtain embeddings using a custom-designed feedforward network that converts the highly dimensional outputs of the ESM-2 model [77] into compact embeddings suitable for predicting DBRs. We combine these low-dimensional embeddings with a comprehensive collection of sequence-derived features, including rel-

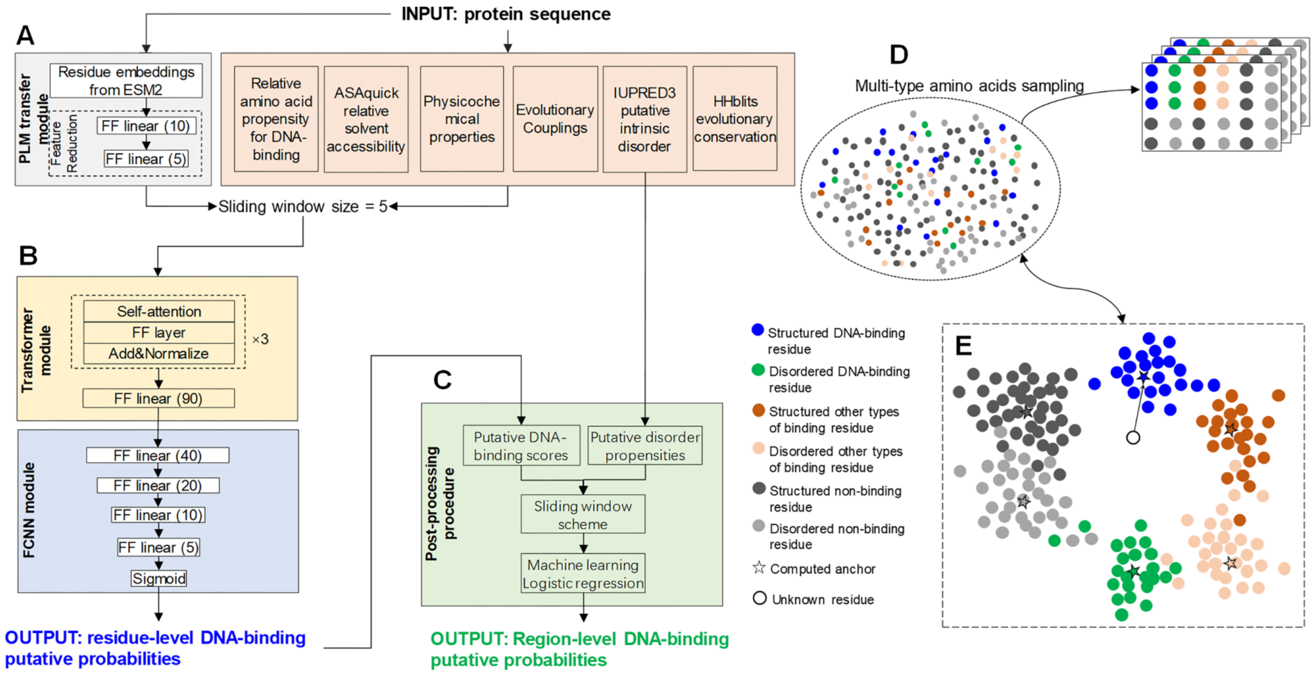

Figure 1. DNAreader's workflow. (**A**) Generation of the sequence profile that includes the PLM module that selects informative embeddings related to the prediction of DBRs from the outputs of the ESM2 model. (**B**) A deep learner that predicts DBRs, which consists of a transformer encoder module and a fully connected feed-forward neural network (FCNN) module. (**C**) The DNAreader$_{\mathrm{DBIDR}}$ model that predicts DNA-binding IDRs. (**D**) Batch training where we designed batches that encompass six representative populations of residues: structured DBRs, structured residues that bind other types of ligands, structured non-binding residues, disordered DBRs, disordered residues that bind other types of ligands, and disordered non-binding residues. (**E**) Contrastive learning, where we train the latent feature space of the FCNN module from panel (B) using the six anchors defined by batch learning, which are denoted by star markers. A test/unlabeled residue is mapped into this feature space by comparing it with the anchor feature spaces.

ative amino acid propensities for binding, relative solvent accessibility, relevant physicochemical properties of amino acids, evolutionary conservation and couplings, and putative intrinsic disorder. Finally, we process this information to generate a sequence profile using a sliding-window approach, which captures the context of neighboring residues to predict the propensity of the residue in the middle of the window to bind DNA. In the second step (Fig. 1B), we feed this profile into the transformer encoder-based predictor, which outputs a real-valued propensity for DNA binding. We produced this network using batch training (Fig. 1D) and contrastive learning with a custom-designed loss function (Fig. 1E) to maximize predictive performance and minimize cross-predictions. We tailored the network training via contrastive learning to reflect our scenario, which includes multiple distinct populations/types of residues, i.e., structured versus disordered, DNA-binding versus other ligand-binding versus non-binding. In the third step (Fig. 1C), we extend our predictor to produce the DNA-binding IDRs. To this end, we trained a secondary model, DNAreader$_{\mathrm{DBIDR}}$, that uses logistic regression to process predictions of DBRs generated in the second step that are combined with putative IDRs generated by the modern IUPred3 method [78] to generate predictions of DNA-binding IDRs. In other words, DNAreader$_{\mathrm{DBIDR}}$ outputs predictions that are compatible with the outputs of the current disorder-trained predictors of DBRs. The following sub-sections provide further details.

Step 1: Generation of the sequence profile

As depicted in Fig. 1A, the sequence profile includes residue embeddings and sequence-derived features processed using a sliding window. The former are computed with the ESM-2 model, a self-attention-based PLM [77], which represents sequence order, and which was recently used in related prediction studies [79, 80]. We utilize the smallest model (esm2_t6_8M_UR50D with 8M parameters) to minimize overfitting and the computational footprint of our predictor. The output of the esm2_t6_8M_UR50D model consists of 320-dimensional residue embeddings, which we feed into a fully connected feed-forward neural network (FCNN) that selects the most informative embeddings for predicting DBRs. Our objective was to reduce the dimensionality of the inputs to the subsequently used transformer encoder-based model. This reduces the size of that transformer encoder network, lowering the risk of overfitting. The FCNN is relatively small, consisting of two layers with 10 and 5 nodes. Each layer is followed by a Rectified Linear Unit (ReLU) activation function, which we used to accelerate the model's convergence during training. We also employed a dropout layer after each activation layer, with a dropout rate of 0.3, to mitigate overfitting. We pre-trained this module using the training dataset 1A and the validation dataset 1, employing the popular Adam optimizer and cross-entropy loss function to differentiate DBRs from non-DBRs. We selected the model that achieved the highest AUROC on validation dataset 1, while keeping the difference between the training and validation AUROCs low (below 0.03) to mitigate potential overfitting. The final model achieved AUROCs of 0.815 and 0.834 on the validation dataset 1 and the training dataset 1A, respectively. Next, we used transfer learning by freezing this model during training of the transformer encoder network shown in Fig. 1B.

The sequence-derived features encompass physicochemical aspects of proteins relevant to predicting DBRs [13], complementing the information extracted by PLM. They include relative amino acid propensity for protein–DNA binding calculated using Composition Profiler [81] based on an approach from Zhang *et al.* [13]; relative solvent accessibility predicted with fast and accurate ASAquick [82]; selected amino acids properties from the AAindex resource [83]: charge, hydrophobicity, polarizability, and electron-ion interaction potential (Supplementary Table S3); putative disorder propensities computed by IUPred3 [78]; and multiple sequence alignment generated by fast HHblits using the relatively small UniProt30 dataset [84], which we utilized to compute evolutionary couplings based on the approach from Zhang *et al.* [76]. We combined the FCNN module outputs with sequence-derived features and used a sliding window of size 5 to generate the sequence profile, yielding 90 features (Supplementary Table S4). We used a relatively short window, which results in a compact input size and, consequently, a smaller transformer network, reducing the risk of overfitting.

Step 2: Design and training of the transformer encoder-based network

We input the sequence profile into a transformer-based network that predicts residue-level DNA-binding propensity. This network consists of three stacked transformer encoder units followed by an FCNN (Fig. 1B). Each unit consists of a multi-head self-attention unit, a feedforward layer, and a normalization layer. The self-attention unit employs a multi-headed attention mechanism to capture dependencies between amino acids along the sequence (i.e., within the sliding window), which are particularly useful for modeling relations related to physicochemical features. The transformer encoder block is followed by a four-layer FCNN composed of ReLU nodes that gradually reduce the latent feature spaces to a single output. The final layer's output node uses the sigmoid activation function, which produces real-valued propensities. We trained the network on the training dataset using the Adam optimizer and used the corresponding validation set to establish a stopping criterion. More specifically, we stop training when the AUROC on the validation set reaches its maximum, while ensuring that the difference in AUROC between the training and validation sets does not exceed 0.03 to prevent overfitting. We also limit the number of training epochs to 100 and note that training converged in 30–80 epochs.

Importantly, we train this network utilizing batch training [69] and contrastive learning [70], which, as we demonstrate in the ablation analysis, substantially boosts predictive performance and reduces cross-predictions. In the mini-batch gradient descent training, each training epoch (iteration) utilizes a small subset of training samples, known as a batch [69]. For protein prediction problems, batches can be generated from a small collection of training sequences or randomly by sampling amino acids from the training dataset. The training process computes predictions for a given batch and calculates the loss, which is used to update the network parameters via gradient computation. This iterative process continues over multiple epochs until the model converges to an "optimal" solution, typically evaluated on a validation dataset. We innovated this approach by ensuring that the batch generation and the network training take advantage of the fact that we identified six distinct types (sub-populations) of residues: structured DBRs, disordered DBRs, structured residues that bind other types of ligands, disordered residues that bind other types of ligands, structured non-binding residues, and disordered non-binding residues; we explained this categorization in the "Introduction" section. As illustrated in Fig. 1D, we designed training batches to ensure that each adequately covers all six residue types, thereby ensuring effective training. We used batches of 500 randomly selected residues, with proportions of structured and disordered non-binding residues, as well as structured and disordered residues that interact with non-DNA ligands, reflecting their native proportions in the training dataset 1B. Since structured and disordered DBRs are in the minority, we fine-tuned their rates by performing a grid search over 1%, 2%, 3%, 4%, and 5% values. This resulted in selecting the 4% rates for disordered DBRs and structured DBRs, which produced the highest AUROC on the validation dataset 1.

Traditional training optimizes neural network models by maximizing their ability to distinguish between labels, in our case, DBRs and non-DBRs. Correspondingly, these networks' latent feature spaces pull objects with the same label closer together while pushing objects with different labels apart. We applied a contrastive learning strategy (Fig. 1E) that extends this principle by utilizing anchors that are used to train latent feature spaces to represent multiple sub-populations (types) of objects [70, 85], which in our case are different types of DBRs and non-DBRs. The underlying ability to organize the latent feature space where objects of the same type are grouped relies on designing an appropriate loss function and adequately representing all residue types in the training batches. We employed multiple anchors to create a multi-center contrastive loss function ($\mathrm{loss_{MC}}$) that minimizes intra-type loss ($\mathrm{loss_{MCintra}}$) while maximizing the inter-type loss ($loss_{MCinter}$) using the 90-neurons latent feature space at the output from the stacked transfer encoder (Fig. 1B), which we denote as $f(x_i)$, as follows:

$$\begin{aligned}\mathrm{loss_{MC}} &= \mathrm{loss_{MCintra}} + \mathrm{loss_{MCinter}}\\ &= \sum_i D\left(f\left(x_i\right), \mathrm{type_{y_i}}\right)\\ &+ \sum_i \sum_{j\neq y_i} \max\Gamma(\Delta_m - \left(D\left(f\left(x_i\right), \mathrm{type_{y_i}}\right), 0\right),\end{aligned}$$

where x_i is the input that is associated with the ith residue type, $type_{yi}$ is the latent feature space of the ith anchor y_i, Δ_m is a margin hyperparameter that defines the minimum distance between centers of different amino acid types (we set it to the default value of 0.3), and $D(\cdot)$ is the squared Euclidean distance function between corresponding latent feature spaces:

$$D\left(f\left(x_i\right), \mathrm{type_{y_i}}\right) = \frac{1}{2}\| f\left(x_i\right) - \mathrm{type_{y_i}}\|_2^2.$$

We combined the multi-center contrastive loss with the "traditional" focal loss (loss_F) [86], which we applied in a recent related study [76] and which depends on the difference between the native binary labels (1 for DBRs versus 0 for non-DBRs) and the network output p:

$$\mathrm{loss_F} = -(1-p)^2 \log\left(p\right).$$

The final loss function is a weighted sum of the contrastive and focal loss elements:

$$\mathrm{loss} = \omega \cdot \mathrm{loss_{MC}} + (1-\omega) \cdot \mathrm{loss_F},$$

where ω is the weight that balances the two types of loss values. We fine-tuned this function by performing a grid search of the ω values in the 0, 0.1, 0.2, ..., 1 set, and selecting the value that produces the highest AUROC on the validation dataset 1.

We considered multiple anchor selections, including three, five, six, and eight anchors, which progressively yield finer, smaller sub-populations. The three anchors cover DBRs, residues that bind all other ligands and non-binding residues. The five anchors include the non-binding residues and divide binding residues into structured and disordered, for both DBRs and residues that interact with other ligand types. The six anchors further divide the non-binding residues into disordered and structured. The eight anchors subdivide the residues that bind non-DNA ligands into those that interact with proteins and those that interact with non-DNA and non-protein ligands, given that the largest group of non-DBRs binds proteins. We trained networks with these four anchor configurations using the same strategy to fine-tune the ω weights by maximizing performance on validation dataset 1, as described above, and report these results in Supplementary Fig. S1. We found that as the number of anchors increases, the predictive performance, as measured by AUROC and F1, improves, while cross-prediction and over-prediction errors, as measured by AUCPC and AUOPC, decrease. More specifically, increasing the number of anchors from three or four to six led to large and statistically significant improvements (P-values $<.01$). However, the differences in AUROC, F1, AUCPC, and AUOPC between six and eight anchors are small and not statistically significant (P-values $>.05$). This means that subdividing the residues that bind non-DNA ligands does not improve performance, suggesting that these subpopulations are either too small or not substantially different from one another. Consequently, we adopted the six-anchor configuration to implement contrastive learning. Empirical optimization of this setup resulted in selecting $\omega = 0.8$, revealing that the primary contributor to the loss calculation is the multi-center contrastive loss.

We also compared the transformer-based design with a convolutional neural network architecture. We constructed both transformer and convolutional encoders using the same input features (Fig. 1A) and similar network sizes, including stacking of three transformer/convolutional units and an FCNN module (Fig. 1B), and trained them using the same batch training (Fig. 1D) and contrastive learning strategies (Fig. 1E). The corresponding side-by-side comparison is shown in Supplementary Fig. S2. The results indicate that the transformer encoder significantly outperforms the convolutional encoder across all metrics (P-values $<.01$), including AUROC and F1, which quantify predictive accuracy, and AUCPC and AUOPC, which quantify cross-prediction and over-prediction errors. These improvements can be attributed to the transformer's ability to capture long-range dependencies among residues and its dynamically generated attention weights. These results motivated our selection of the transformer architecture to implement DNAreader.

Step 3: Design and training of the DNAreader$_{\mathrm{DBIDR}}$ module

Besides predicting DBRs in the structured and disordered regions, DNAreader also provides predictions of DNA-binding IDRs via the DNAreader$_{\mathrm{DBIDR}}$ module (Fig. 1C). This additional (secondary) prediction mimics the results produced by the current disorder-trained predictors. This module aims to identify IDRs predicted to have DBRs and, correspondingly, relies on two inputs: putative DBR propensities generated by the transformer-based module of DNAreader and IDRs predicted from the input protein sequence with the modern IUPred3 tool [78]. We used these inputs to derive 33 numerical features, which are summarized in Supplementary Table S5. Given that the aim is to predict disordered binding regions, several features aggregate intrinsic disorder predictions (averages and standard deviations) over sliding windows of various sizes. We used a logistic regressor as the predictor since the underlying patterns should be simple (binding IDRs are essentially the input IDRs with the input predicted DBRs), and this model is unlikely to overfit the training data [87]. We trained the regression model on the training dataset 2 and optimized it to maximize performance on the validation dataset 2, i.e. we selected the model with the highest AUROC value on the validation set 2, while ensuring that the difference in AUROC between the training dataset 2 and validation datasets 2 is <0.03; the latter aims to mitigate the risk of overfitting.

Results and discussions

Ablation analysis

DNAreader relies on two innovations: contrastive learning (a custom-designed and empirically optimized loss function) and custom-crafted batch training, which together aim to boost predictive performance and reduce cross-predictions. We empirically assess the impact of these innovations by performing ablation analysis where we compare the DNAreader with its variants that include removing batch training and instead relying on the commonly used sliding window-based training (Model 1); removing contrastive learning and instead performing traditional learning with the focal loss function (Model 2); and removing both contrastive learning and batch training (Model 3).

Figure 2A compares the predictive performance of DNAreader against the three ablation setups on test set 1. We found that DNAreader is significantly more accurate than each of the three ablation variants across four metrics that evaluate overall predictive performance and the cross-predictions and over-predictions: AUROC, F1, AUCPC, and AUOPC (P-values $<<.01$). Removing both innovations resulted in a large drop in AUROC from 0.863 to 0.803, accompanied by a substantial decline in cross-predictions, with an increase in AUCPC from 0.121 to 0.172. Moreover, each innovation contributes to these improvements. While, in relative terms, removing batch training (Model 1) produces a smaller negative impact than removing contrastive learning (Model 2), the predictive performance is consistently and significantly lower in both cases (P-values $<<.01$). We particularly noted the large increase in AUCPC when either innovation is removed individually, from 0.121 to 0.158 (Model 1) and 0.168 (Model 2). We also analyzed these ablation results separately for the disordered-annotated and structure-annotated proteins in test set 1, as illustrated in Supplementary Figs S3 and S4, respectively. These results produce the same conclusions as Fig. 2A, revealing that the two innovations significantly improve both types of annotations.

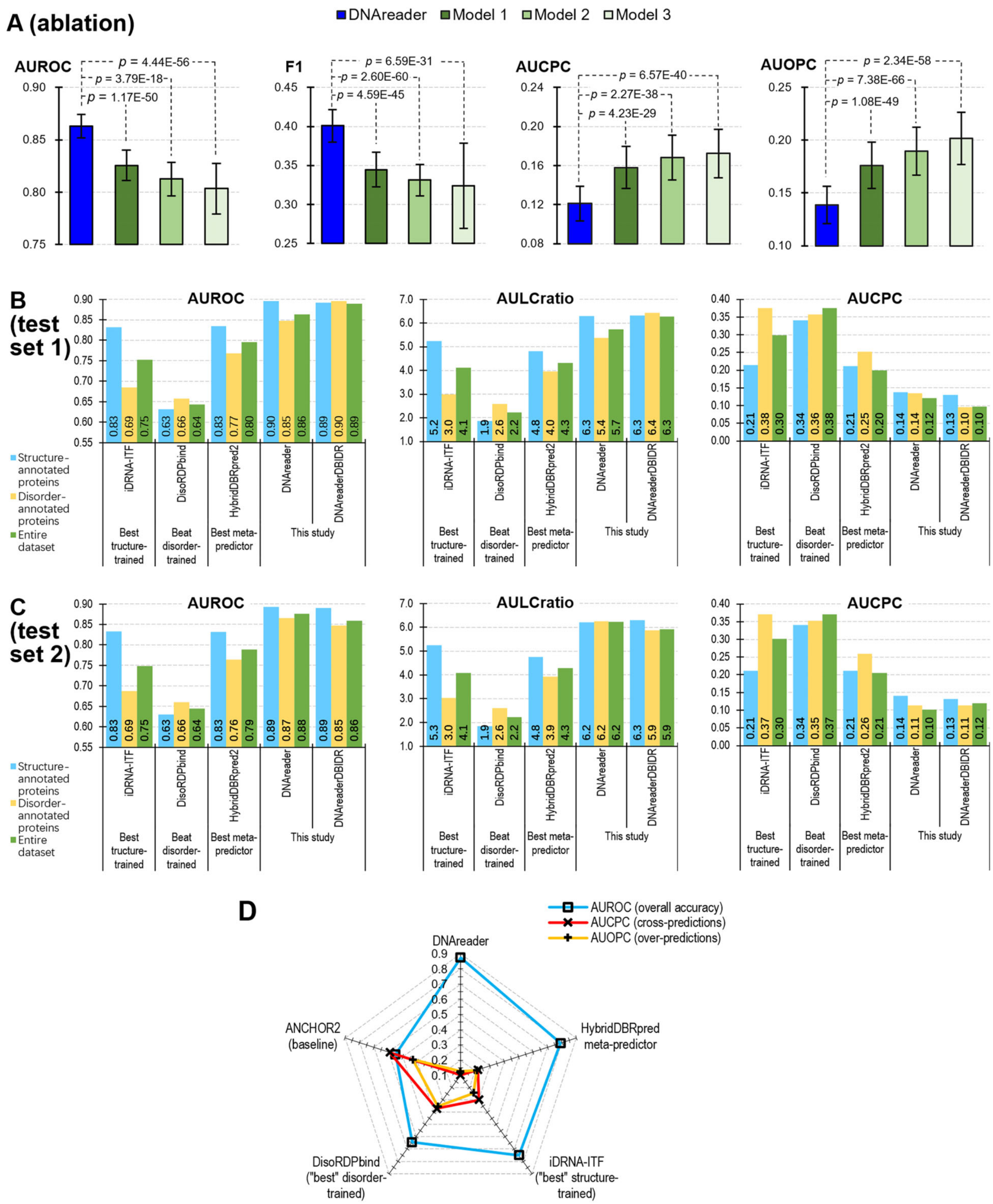

Figure 2. Empirical comparative assessment. (**A**) Ablation analysis on the test dataset 1 that compares DNAreader with its variants that remove batch training (Model 1), delete contrastive learning (Model 2), and remove both contrastive learning and batch training (Model 3). We report AUROC and F1, which evaluate predictions of real-valued propensities and binary predictions, respectively, and AUCPC and AUOPC, which evaluate cross-predictions and over-predictions, respectively. We show the *P*-values that assess the significance of differences between DNAreader and its variants above the bars. Comparison of the predictive performance on test set 1 (**B**) and test set 2 (**C**) for the selection of the best tools, including the best structure-trained predictor, the best disorder-trained predictor, the best meta-predictor, and DNAreader. We report AUROC and AULCratio, which evaluate predictive performance, and AUCPC, which evaluates cross-predictions, on the complete dataset (green bars), the subset of structure-annotated proteins (blue bars), and the subset of disorder-annotated proteins (orange bars). The full assessment results covering all predictors are in Supplementary Tables S6, S7, and S8, and include the results of the statistical significance analysis. (**D**) Side-by-side comparison of DNAreader against the best current tools that include the HybridDBRpred meta-predictor, structure-trained iDRNA-ITF, disorder-trained DisoRDPbind, and the baseline ANCHOR2 that predicts ligand-type agnostic binding residues on the test dataset 2. We detailed the assessment metrics and tests used in these panels in the "Assessment metrics and statistical analysis" section.

Comparative assessment on test dataset 1 using region-level annotations for the disorder-annotated interactions

This evaluation relies on the blind (low-similarity) test dataset 1 that we derived from our recently published benchmark set [61]. This dataset includes DNA-binding annotations consistent with those used in related studies, specifically DBRs for structure-annotated proteins and DNA-binding IDRs for disorder-annotated proteins. We assessed DNAreader and its $\mathrm{DNAreader_{DBIDR}}$ variant against 14 representative predictors, including 8 structure-trained predictors, 2 disorder-trained predictors, 3 meta-predictors, and the ANCHOR2 baseline. Furthermore, we analyzed the complete test set and the subsets of disorder-annotated and structure-annotated proteins in this dataset. We provided these results in Supplementary Tables S6 (overall predictive performance) and Supplementary Table S7 (cross-predictions and over-predictions), with the corresponding ROC curves in Supplementary Figs S5, S6, and S7. We visualize a key subset of these results for the best structure-trained predictor, the best disorder-trained predictor, the best meta-predictor, and DNAreader, in Fig. 2B.

Evaluation of the overall predictive performance

Supplementary Table S6 reveals that the best structure-trained methods produce very accurate predictions for the structure-annotated proteins, achieving AUROCs >0.80 and AULCratios >5, with the highest values of 0.83 and 5.3 for iDRNA-ITF (Fig. 2B), respectively, while they perform less well for the disorder-annotated proteins, with AUROCs ≤0.70 and AULCratios <3.2. The disorder-trained tools underperform for the structure-annotated proteins, with AUROCs ≤0.63 and AULCratios ≤1.9, whereas they generate relatively more accurate results for the disorder-annotated proteins, with an AUROC of 0.66 and an AULCratio of 2.6 for DisoRDPbind (Fig. 2B). This dichotomy, where the disorder-trained methods relatively underachieve for the structure-annotated proteins, and the structure-trained methods underperform for the disorder-annotated proteins, is in line with observations in recent studies [61, 64] and underlines the need to develop methods that cross over between the two annotation types. Our analysis shows that the first of these attempts, which produced the HybridDBRpred meta-predictor [61], resulted in noticeable improvements, matching the performance of the best structure-trained method on the structure-annotated proteins (AUROC of 0.82 and AULCratio of 4.4), and improving over both structure- and disorder-trained methods for the disorder-annotated proteins (AUROC of 0.77 and AULCratio of 4.0). Interestingly, we observed that three recently released structure-trained DNAgenie, CLAPE, and iDRNA-ITF secure AUROCs around 0.68 and AULCratios ranging from 2.4 to 3.2 for the disorder-annotated proteins, matching the results of the best disorder-trained method. This stems from advancements utilized in their predictive models, such as the use of disorder predictions and a well-optimized support vector machine model in DNAgenie [45], and the use of advanced deep neural network architectures and PLM in CLAPE [55] and iDRNA-ITF [47]. However, we emphasize that these three tools still underperform for the disorder-annotated proteins compared with their predictions for the structure-annotated proteins. However, the use of these newest tools to develop a new variant of the HybridDBRpred meta-predictor, HybridDBRpred2, did not result in further improvements compared to the original meta-predictor; AUROC of 0.82 versus 0.83 and AULCratio of 4.4 versus 4.8 for the structure-annotated proteins, and AUROC of 0.77 and AULCratio of 4.0 for both variants of HybridDBRpred for the structure-annotated proteins. This limitation in the predictive performance and the need to predict DBRs for IDRs motivated the development of DNAreader.

The evaluation on the entire test dataset 1 in Supplementary Table S6 yielded several informative insights. Among the eight structure-trained methods, iDRNA-ITF and CLAPE achieve the best performance, with AUROCs of 0.75 and 0.73, respectively, and AULCratios of 4.1 and 3.7. DisoRDPbind outperforms the other disorder-trained methods, attaining an AUROC of 0.64 and an AULCratio of 2.2. The HybridDBRpred and HybridDBRpred2 meta-predictors achieve AUROCs of ∼0.79 and an AULCratio of ∼4.3. More importantly, DNAreader outperforms the best structure-trained, disorder-trained, and meta-predictors by a significant margin (P-values <.01), achieving an AUROC of 0.86 and AULCratio of 5.7 on the entire dataset, AUROC of 0.90 and AULCratio of 6.3 on structure-annotated proteins, and AUROC of 0.85 and AULCratio of 5.4 on disorder-annotated proteins (Fig. 2B). The improvements offered by DNAreader are similarly large and statistically significant (P-values <.01) for the other metrics. For instance, on the full dataset, DNAreader has a sensitivity of 0.61 at a low FPR of 0.1, a specificity of 0.94 at a TPR of 0.5, and an F1 score of 0.40. Figure 2B and Supplementary Table S6 reveal that $\mathrm{DNAreader_{DBIDR}}$ secures similar results to DNAreader on the structure-annotated proteins while generating significantly more accurate results for the disorder-annotated proteins (AUROC of 0.90 versus 0.85 and AULCratio of 6.4 versus 5.4; P-value <.01), consequently producing more accurate predictions on the entire test dataset 1, with AUROC of 0.89 and AULCratio of 0.63 (P-value <.01). This is expected since the $\mathrm{DNAreader_{DBIDR}}$ module augments DNAreader's predictions of DBRs in the disorder-annotated proteins into the disordered region-level, resulting in predictions more compatible with the annotations used in the test dataset 1. At the same time, it aims not to adjust the DBR predictions for the structure-annotated proteins.

Altogether, our empirical test demonstrated that DNAreader produces highly accurate predictions of DBRs, providing substantial, statistically significant improvements over current tools. Furthermore, its $\mathrm{DNAreader_{DBIDR}}$ variant offers additional improvements in predicting DNA-binding IDRs.

Evaluation of cross-predictions and over-predictions

DNAreader and the other predictors should selectively identify binding residues that interact with DNA. However, several studies have shown that some of these methods are ligand-type agnostic, i.e., they tend to predict residues that interact with other ligand types as DNA-binding at similar rates that they predict the native DBRs [47, 52, 61, 64]. To this end, we comparatively evaluated cross-predictions (incorrect predictions of DBRs among residues that interact with non-DNA ligands) with AUCPC (lower is better) and CPRratio (higher is better), and over-predictions (incorrect predictions of DBRs among residues that do not interact with ligands) with AUOPC (lower is better) and OPRratio (higher is better); details are in the "Assessment metrics and statistical analysis" section. Supplementary Table S7 provides these results, while Fig. 2B visualizes the AUCPC values for the best structure-trained predictor, the best disorder-trained predictor, the best meta-predictor, and DNAreader. The complete sets of cross-prediction and over-prediction curves are available in Supplementary Figs S6 and S7.

Most structure-trained and disorder-trained methods are characterized by high cross-prediction rates, with CPRratio <2 or AUCPC ≥0.35, which suggests that they are ligand-type agnostic (Supplementary Table S7). These include the structure-trained TargetS, TargetDNA, BindN+, and DNAPred, as well as the disorder-trained DeepDISOBind and DisoRDPbind. DNAgenie, iDRNA-ITF, CLAPE, and NCBRPred produce modest levels of cross-predictions on the test set 1 (CPRratio between 2 and 2.5, and AUCPC between 0.25 and 0.35). The meta-predictors improve over the disorder-trained and structure-trained methods across structure-annotated and disorder-annotated proteins, securing the overall CPRratio around 3 and AUCPC around 0.20. Importantly, DNAreader generates significantly fewer cross-predictions than the other predictors, including the meta-predictors, for structure- and disorder-annotated proteins (p-values < 0.01), thereby securing the overall CPRratio > 4.5 and AUCPC < 0.13 on test set 1 (Fig. 2B and Supplementary Table S7). This implies that DNAreader specifically predicts DBRs, with very low cross-prediction rates. Similar to what we observed in Supplementary Table S6, Supplementary Table S7 showed that $DNAreader_{DBIDR}$ performs significantly better than DNAreader on the disorder-annotated proteins (*P*-values <.01) and performs similarly well on the structure-annotated proteins, with an overall CPRratio of 4.8 and an AUCPC of 0.10.

We also analyzed the over-predictions in Supplementary Table S7. Like for the cross-predictions, individual structure- and disorder-trained predictors are surpassed by the meta-predictors, which in turn are significantly outperformed by DNAreader in OPRratio and AUOPC on both structure-annotated and disorder-annotated proteins (*P*-value <.01). DNAreader and $DNAreader_{DBIDR}$ obtain the overall OPRratios of 3.8 and 4.1, and AUOPC of 0.14 and 0.11 on the entire test set 1. These modestly better results from $DNAreader_{DBIDR}$ can be attributed to its much better performance on disorder-annotated proteins (Supplementary Table S7). Moreover, the results on the entire test set 1 showed that the structure- and disorder-trained tools consistently generate much higher cross-predictions rates than over-predictions rates (CPRratio lower than OPRratio, AUCPC higher than AUOPC). This is because residues that bind the other ligand types are more similar to the DBRs than the residues that do not interact with ligands, and since most of these tools were not designed to prevent cross-predictions (Supplementary Table S1). However, this trend does not appear for the meta-predictors and DNAreader, which have been specifically designed to minimize the cross-predictions.

Overall, consistent with the analysis in the "Evaluation of the overall predictive performance" section, we observed that DNAreader is more accurate than the current best tools by large and statistically significant margins (Fig. 2B). Furthermore, $DNAreader_{DBIDR}$ variant outperforms DNAreader, which suggests that $DNAreader_{DBIDR}$ successfully extends the DNAreader's predictions of DBRs into the DNA-binding IDRs, given that such regions constitute the ground truth in this test set for the disorder-annotated proteins.

Comparative assessment on the test dataset 2 using DBR annotations for the disorder-annotated and structure-annotated interactions

We also evaluated DNAreader and the other tools on the new blind (low-similarity) test dataset 2, which uses consistent annotations of DBRs for the structured and disordered test proteins; see Supplementary Table S8. We summarize these results for the best structure-trained predictor, the best disorder-trained predictor, the best meta-predictor, and DNAreader in Fig. 2C. These results underpin the key motivation of this study: the development of the first tool that was specifically designed to predict DBRs for the two protein groups.

We observed consistent trends within the structure-disorder dichotomy. The structure-trained methods generated a higher average (over the eight structure-trained methods) AUROC of 0.77 and an average AULCratio of 3.9 for the structure-annotated proteins versus a much lower average AUROC of 0.61 and an average AULCratio of 2.3 for the disorder-annotated proteins (Supplementary Table S8). Similarly, the disorder-trained methods achieved an average AUROC of 0.57 and an average AULCratio of 0.84 for the structure-annotated proteins, versus higher average values of 0.61 and 2.0 for the disorder-annotated proteins in test set 2. On the other hand, the HybridDBRpred meta-predictor obtained more consistent and accurate results for the two protein groups, with an AUROC of 0.83 versus 0.76 and an AULCratio of 4.4 versus 3.9.

Figure 2C shows that DNAreader generated very accurate predictions, with an AULCratio of about 6.2 and an AUROC of 0.89, 0.86, and 0.88 for the structure-annotated, disorder-annotated, and all proteins on test set 2. It significantly improves over its $DNAreader_{DBIDR}$ variant that secured the overall AULCratio of 6.0 and AUROC of 0.86 (*P*-value <.01; Supplementary Table S8). This is because $DNAreader_{DBIDR}$ aims to predict DNA-binding IDRs rather than DBRs in the disorder-annotated proteins, and it correspondingly obtained worse results on these proteins. Similar to the results in the "Comparative assessment on test dataset 1 using region-level annotations for the disorder-annotated interactions" section, we observed that all other methods are significantly less accurate than DNAreader (*P*-values <.01), with an AULCratio of 4.3 and an AUROC of 0.79 for the best meta-predictor, HybridDBRpred2, and an AULCratio of 4.1 and an AUROC of 0.75 for the best individual structure- or disorder-trained predictor, iDRNA-ITF (Fig. 2C). The evaluation of cross-predictions and over-predictions leads to the same observations. DNAreader obtained low values of AUCPC of 0.10 and AUOPC of 0.13 compared to statistically worse results of the other tools (*P*-values <.01), with an AUCPC of 0.20 and an AUOPC of 0.21 for HybridDBRpred2, and an AUCPC of 0.30 and an AUOPC of 0.24 for iDRNA-ITF (Supplementary Table S8 and Fig. 2C).

Our empirical assessment on test set 2 revealed that DNAreader accurately predicts DBRs in structured and disordered sequence regions, consistently generates low rates of cross-predictions and over-predictions, and outperforms the currently best methods by large, statistically significant margins.

Case studies

We use two distinct test proteins to illustrate and explain predictions from DNAreader and to contrast them with the second-best method, the HybridDBRpred2 meta-predictor. The first protein, human estrogen-related receptor-2 (ERRβ; UniProt ID: O95718), binds DNA and demonstrates how to interpret DNAreader's predictions of DBRs and DNA-binding regions. The second protein, L-lactate 2-monooxygenase from *Mycolicibacterium smegmatis* (UniProt ID: P21795), does not interact with DNA but binds two other ligands, flavin mononucleotide (FMN) and 2-oxocarboxylate, showing how DNAreader handles cross-predictions. We run DNAreader, $\text{DNAreader}_{\text{DBIDR}}$, and hybridDBRpred2 on these two protein sequences and compare the resulting predictions with the native annotations of binding regions and residues. Fig. 3A and B visualize the predicted scores for three methods along the protein sequence using the color-coded line plots (dark blue for DNAreader; light blue for $\text{DNAreader}_{\text{DBIDR}}$, and green for hybridDBRpred2). The scores are normalized to the 0–1 range, where higher scores indicate a greater propensity for the corresponding residues to bind DNA. We also computed binary predictions from these scores using thresholds corresponding to a low false-positive rate of 0.1 on test dataset 2 (residues with scores above the threshold are predicted as DBRs), consistent with the experimental setup described in the "Assessment metrics and statistical analysis" section. We illustrate these thresholds using horizontal dashed lines. The corresponding putative binding residues are at the bottom, as color-coded diamond markers. Depending on the user's tolerance for false positives, these thresholds can be adjusted; lower thresholds yield more binding residues but higher false-positive rates. Residues with scores much higher (or lower) than the threshold are more likely to be DBRs (non-DBRs) when compared with residues with scores closer to the threshold.

The ERRβ protein is a 433-amino-acid nuclear receptor that is structurally and functionally similar to estrogen receptors, but it does not bind endogenous estrogen [88]. Instead, ERRβ interacts with several nuclear receptors [89, 90] and regulates genes involved in multiple metabolic processes [91]. An impaired ERRβ gene was shown to lead to increased fetal lethality [92], while its overexpression was reported to inhibit cell proliferation in breast cancer [93, 94] and prostate cancer [95]. This protein has a conserved DNA-binding domain (DBD) with two zinc-finger motifs that bind specific DNA sequences in the promoter regions of target genes [96]. The DBD extends from residue 97–194, where the core region with a structured and well-defined zinc-finger motif ends at position 168, and the C-terminal extension hosts a disordered region from position 187 to 194. We show the corresponding native annotations of DNA-binding regions and DBRs, which span the structured and disordered sequence segments of the DBD, using black and gray markers, respectively, positioned just below the x-axis, which denotes the sequence (Fig. 3A). Scores produced by hybridDBRpred2 (green markers in Fig. 3A) have higher values for the entire DBD, and correspondingly, they cannot be used to pinpoint the location of native DBRs. They also drop substantially for the DBRs in the disordered segment of the DBD (positions 187–194), thereby missing this DNA-interacting region. $\text{DNAreader}_{\text{DBIDR}}$ (light blue markers in Fig. 3A) produces high scores (i.e., scores substantially larger than the threshold) for residues in structured and disordered DNA-binding regions. However, when score values are close to the threshold, it fails to predict DBRs at positions 150–160. This can be explained by the fact that this model was trained to predict binding regions rather than residues; consequently, it may miss shorter binding regions or more isolated DBRs. DNAreader produces high scores for the DBRs across both structured and disordered DNA-binding regions. The four residues that DNAreader incorrectly predicts as DBRs at positions 287, 288, 405, and 409 have scores in the moderate (near-threshold) range. Correspondingly, they should be considered spurious (false positive) predictions, which would be removed using a more stringent (higher) threshold that produces a lower false positive rate. Figure 3C visualizes the sequence-based DNAreader's predictions on the experimental 3D structure of the DBD of ERRβ complexed with DNA (PDB ID: 1LO1). The protein chain is colored to reflect the DNA-binding propensities predicted by DNAreader, which range from red (low scores denoting likely non-binding residues), through white (moderate scores near the threshold with no apparent bias toward either DNA-binding or non-binding), to blue (high scores for likely DBRs). Native DBRs are shown as spheres, while the rest of the protein is shown in the cartoon format. The structured DNA-binding region forms an α-helix, while the disordered region (marked with black ovals in Fig. 3B) is composed of flexible loops and turns. The figure shows that the native DBRs are predominantly colored dark blue, indicating high confidence that they are correctly predicted as DBRs. Moreover, the 180° rotated view of the complex (the right sub-panel in Fig. 3C) shows that for the right-most helix, the binding residues are colored in blue, while the rest of the residues are either white or red, suggesting that DNAreader correctly identifies them as less likely to interact with DNA. Altogether, this case study reveals that DNAreader accurately reproduced the amino-acid-level details of protein–DNA interactions from the protein sequence.

The L-lactate 2-monooxygenase is an enzyme composed of 394 amino acids that catalyzes the oxidative decarboxylation of hydrogen peroxide, producing acetate, carbon dioxide, and water [97]. This is a structured protein that interacts with FMN and 2-oxocarboxylate ligands, with no evidence of DNA-binding. Ideally, the considered predictors should yield propensities near 0 for all amino acids in this protein. Figure 3B shows that scores produced by DNAreader are generally much lower than the threshold, except for just six residues, where scores rise slightly above it. However, these scores fall within the moderate (near-threshold) range, suggesting that there are likely false positives. Altogether, these predictions reveal that DNAreader correctly identifies this protein as not interacting with DNA. The results from the $\text{DNAreader}_{\text{DBIDR}}$ variant are even more accurate, as its predicted scores are mostly below 0.1 and consistently much lower than the threshold. This can be explained by the fact that this is a structured protein, and correspondingly, $\text{DNAreader}_{\text{DBIDR}}$, which is designed to predict disordered DNA-binding regions, does not identify either DBRs or disordered regions. In contrast, hybridDBRpred2 predicts 30 residues with scores above its threshold, including 5 cross-predictions located where this protein binds non-DNA ligands, and 25 over-prediction errors. Notably, some of these predicted scores substantially exceed the threshold, particularly in positions 43–55 and 290–303, where cross-predictions occur, suggesting that these incorrect DBR predictions are accurate. This example illustrates the underlying hybridDBRpred2's tendency toward cross-predictions, while showing that DNAreader is capable of avoiding this pattern.

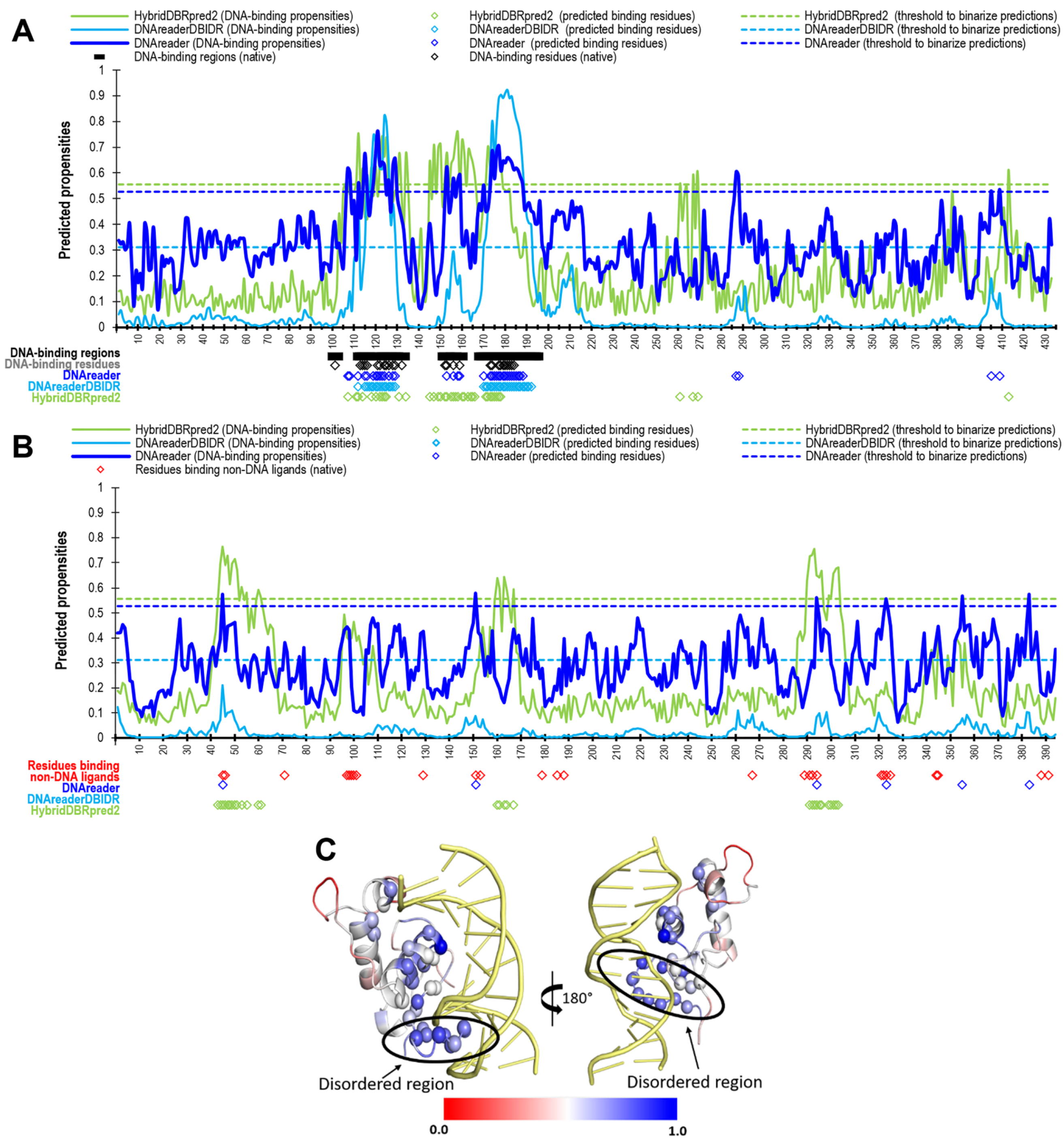

Figure 3. Case study predictions generated from protein sequences. Panels (A) and (B) include color-coded predictions from DNAreader (dark blue), DNAreader$_{DBIDR}$ (light blue), and hybridDBRpred2 (green) along the sequences represented on the *x*-axis. (**A**) Human ERRβ protein (UniProt ID: O95718) that binds DNA,where the native DNA-binding regions and residues are in black and gray at the bottom of the panel. (**B**) L-lactate 2-monooxygenase protein from *M. smegmatis* (UniProt ID: P21795) that binds non-DNA ligands, where the native FMN- and 2-oxocarboxylate-binding residues are in red at the bottom of the panel. (**C**) Visualization of DNAreader's predictions on the structure of the DNA-binding domain of ERRβ covering positions 97 to 190 (PDB ID: 1LO1). The predictions are color-coded according to the scale at the bottom of the panel: low scores are in red (denoting likely non-binding residues), moderate scores are in white (residues with no apparent bias toward either DNA-binding or non-binding), and high scores are in blue (likely DBRs). Native DBRs are shown as spheres, while the rest of the protein is in the cartoon format. DNA is shown in gold.

Availability

We provide the source code for DNAreader at https://github.com/jianzhang-xynu/DNAreader. To boost impact [98], we also implemented DNAreader as a convenient and publicly available web server at http://biomine.cs.vcu.edu/servers/DNAreader/. This web server requires only a FASTA-formatted protein sequence as input. The entire prediction process is automated and performed on the server side, with no need to install additional software on the user's side. Upon submitting their protein sequence, users are redirected to a page that provides updates on the prediction's progress. They then reach the results page. This page provides access to a color-coded visualization of results, including propensities for DNA binding and binary predictions of DBRs for the DNAreader and its $\text{DNAreader}_{\text{DBIDR}}$ variant. Additionally, a comma-separated text file is available for parsing the raw predictions. We advise users to rely on DNAreader's predictions and to use $\text{DNAreader}_{\text{DBIDR's}}$ results only when predicting DNA-binding IDRs. We will retain data from the submissions for at least three months, allowing users to access their previous submissions using the job identifier.

Summary and discussion

Prediction of DBRs in protein sequences has a long history that has resulted in the release of several dozen predictors [8–11], as we demonstrate in Supplementary Table S1. The two key challenges with the current tools are the disorder versus structure dichotomy, where disorder-trained methods underperform on structure-annotated proteins and vice versa, and the lack of tools to predict DBRs in disordered regions. Instead, the current disorder-trained tools were designed to produce less precise predictions of the DNA-binding IDRs. Additionally, many current tools heavily cross-predict, effectively rendering their predictions ligand-type-agnostic.

Our empirical results quantified and confirmed this dichotomy on both test datasets for a representative collection of over a dozen predictors. We addressed these challenges by conceptualizing, implementing, evaluating, and deploying DNAreader, the first and accurate predictor designed to predict DBRs in disordered and structured regions of protein sequences. DNAreader's model uses stacked transformer encoder networks and a modern PLM, leveraging two innovations: custom-crafted batch training and contrastive learning. Our ablation study demonstrated that these innovations yield substantial gains in predictive performance and significant reductions in cross-prediction. Moreover, we implemented several mechanisms to minimize the likelihood of overfitting, including using a small network input and a correspondingly small network size, as well as dropout and low-similarity validation data during network training.

Using two low-similarity test datasets, we demonstrated that DNAreader outperforms the published tools by large and statistically significant margins. A radar plot in Fig. 2D summarizes these results by comparing DNAreader against the HybridDBRpred meta-predictor and the best structure-trained and disorder-trained tools. The blue plots compare overall performance measured by AUROC, while the red and orange plots visualize errors from cross-predictions and over-predictions, respectively. The size of the gap between the blue and red/orange plots indicates that a given method maximizes the accuracy of predicting DBRs and minimizes cross- and over-prediction errors. DNAreader's gap is by far the widest, resulting from securing a very high AUROC (0.88 for DNAreader versus 0.79 for the second-best HybridDBRpred) and low AUCPC and AUOPC values (0.10 and 0.13 for DNAreader versus 0.22 and 0.21 for the second-best HybridDBRpred, respectively). We particularly emphasize the much higher cross-prediction errors for the meta-predictor and the disorder/structure-trained tools. We also used the popular ANCHOR2 method, which predicts protein and peptide binding residues, as a baseline to demonstrate the performance of a method that predicts residues that bind molecules other than DNA. As expected, its cross-prediction errors match the performance of the "correct" predictions, i.e., AUROC and AUCPC are similarly bad, corresponding to random levels of predictive performance.

Moreover, we developed and released a $\text{DNAreader}_{\text{DBIDR}}$ variant of DNAreader, which accurately predicts DNA-binding IDRs. This provides end-users with the flexibility to use DNAreader to identify DBRs in IDRs or to predict entire DNA-binding regions, as current tools do. Our empirical comparative analysis shows that DNAreader outperforms $\text{DNAreader}_{\text{DBIDR}}$ for the predictions of DBRs (test dataset 2) while $\text{DNAreader}_{\text{DBIDR}}$ is better for predicting DNA-binding IDRs (test dataset 1). To enhance their impact, we released DNAreader and $\text{DNAreader}_{\text{DBIDR}}$ as a user-friendly, freely available web server at http://biomine.cs.vcu.edu/servers/DNAreader/. The server page provides access to the datasets developed in this study, which the community can use to further expand this active research area by enabling reliable training and comparative evaluation of future predictive models.

To summarize, DNAreader uses an innovative approach to generate highly accurate predictions of DBRs across both structured and disordered regions, with few cross-predictions. We note that annotations of intrinsic disorder rely on diverse experimental methods and manual curation [75] and are context-dependent, which can lead to inconsistencies. These issues can introduce false positives and false negatives at levels perhaps higher than those in structure-based annotations, which utilize more robust experimental tools and avoid manual curation. This means that while DNAreader shows promising results on both structure- and disorder-annotated proteins, these results should not be directly compared or equated since the quality of the ground-truth information is likely different.

We are considering several avenues for future research. One option is to further improve the DNAreader's predictive performance, for instance, by developing and using a specialized PLM for DNA-binding proteins, which would complement the embeddings from the ESM-2 PLM we currently use as inputs. Similar specialized PLMs were already developed for other protein populations, including DR-BERT for disordered proteins [99], ProstT5 for protein structures [100], and BCRInsight for antibodies [101]. Another potential future work is to pre-compute and make available a large collection of DNAreader's predictions, thereby streamlining access and boosting the impact of our tool. These predictions could be incorporated into the existing, large database of residue-level protein predictions, such as DescribePROT [102, 103] and MobiDB [104, 105]. We are also planning to investigate whether the predictive architecture and specialized training developed for DNAreader would provide similarly good results for predicting interactions with other ligands that suffer a similar disorder-structure dichotomy, such as RNAs, proteins, and peptides.

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Supplementary data

Supplementary data is available at NAR online.

Conflict of interest

None declared.

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Data availability

We provide the source code for DNAreader at https://github.com/jianzhang-xynu/DNAreader and https://doi.org/10.5281/zenodo.21261184. We also implemented DNAreader as a convenient and publicly available web server at http://biomine.cs.vcu.edu/servers/DNAreader/.

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